

SMART INDIA HACKATHON 2024 / 2025

HEALTHCARE & MEDTECH CATEGORY

PROBLEM STATEMENT ID: SIH-MED-2024

Apon Mon (আপোন মন)

আপোনাৰ চিনাকি মনৰ সংগী — Cognitive Companion

Culturally Familiar, Offline-First Cognitive Engagement, Dignified Memory Anchors, and Local Care Tracking for Older Adults & ASHA Workers in North East India.

CORE ENGINE

28 Localized Games

Assamese & English Multi-Stage

ARCHITECTURE

100% Offline-First

PWA + Local IndexedDB Queue

AI COMPANION

3D VRM & Edge AI

Faster Whisper + Gemini 3.1

COMMUNITY IMPACT

ASHA Multi-Patient Hub

Explainable Gap Map Analytics

Team: Visionaries (Team Apon Mon)

Target Region: Assam & North East India

Stack: React 18 · Three.js · FastAPI · Gemini · Faster Whisper

The Rural Dementia Challenge: Alienation & Lack of Tools

Over 8.8 million Indians live with dementia, yet 90% in rural regions remain undetected and unsupported.

BARRIER 1: CULTURAL & LINGUISTIC

Western & Abstract Games

- ▶ Existing cognitive apps rely on sterile shapes, neon timers, and English text.
- ▶ Older rural adults experience intimidation, agitation, and immediate abandonment.
- ▶ No integration of regional heritage (Bihu, Majuli pottery, Loktak weaving, Hornbill).

Consequence: High cognitive resistance and non-compliance.

BARRIER 2: CONNECTIVITY & DEVICES

Unreliable Internet & Cloud Lock

- ▶ North East hilly & riverine terrains have frequent network dropouts.
- ▶ Cloud-heavy apps fail entirely during power cuts or remote village field visits.
- ▶ High server subscription costs exclude rural primary healthcare centers.

Consequence: Zero accessibility in remote ASHA circles.

BARRIER 3: DIAGNOSTIC STIGMA

Caregiver Burden & Stigma

- ▶ Clinical questionnaires induce fear of "failing a mental test".
- ▶ Caregivers lack structured, dignified ways to stimulate loved ones daily.
- ▶ No explainable local tracking to assist doctors during periodic reviews.

Consequence: Delayed intervention & severe caregiver burnout.

Apon Mon: A Dignified, Culturally-Rooted Ecosystem

Transforming cognitive exercise from a stressful clinical chore into a peaceful, celebratory daily ritual.

01 · COGNITIVE GAMES

28 Cultural Games

10 categories spanning Mild, Moderate, and Severe stages. Featuring Assamese tea gardens, Bihu instruments, and NE wildlife.

✓ Zero Failure / Calm Retries

02 · REMINISCENCE

Dignified Anchors

Private family photo albums with voice recordings stored in browser IndexedDB. Preserves core personal identity offline.

✓ Zero Cloud Photo Leakage

03 · REWARD SYSTEM

Memory Garden

Positive reinforcement loop: stars earned from games are spent watering traditional tea bushes, orchids, and Brahma Kamal.

✓ Soothing Web Audio Synthesizer

04 · CLINICAL SUPPORT

ASHA Gap Map

Multi-patient switcher with PIN security ('1234'). Displays Memory Gap Map, 5-game baselines, and one-click PDF clinical export.

✓ Outbox Auto-Sync Queue

Design Ethos: WCAG 2.2 AA accessibility, large 64px touch targets, high contrast, Assamese voice read-aloud, and zero competitive pressure.

ZERO-COST OFFLINE CORE

System Architecture: Multi-Tier Edge & Offline Pipeline

Complete decoupling of offline client logic from optional edge AI enhancement microservices.

Architecture Pillars

- ▶ **Layer 1** React 18 + Vite PWA + Service Worker cache (UI Shell): for zero-bandwidth asset delivery.
- ▶ **Layer 2** `platform.js` manages localStorage (State Engine): attempts, settings, and dynamic adaptive levels.
- ▶ **Layer 3** `mediaStore.js` stores encrypted Blob (Private Media): photos and voice clips in IndexedDB.
- ▶ **Layer 4** `syncQueue.js` maintains an offline outbox (Sync Hub): queue with auto-reconnection flushing.
- ▶ **Layer 5 (AI Service):** FastAPI microservice hosting Faster Whisper STT & Gemini 3.1 Flash-Lite.

End-to-End Data & Execution Flow



Adaptive Engine & Memory Gap Map Analytics

Dynamic difficulty calibration preserving dignity while mapping longitudinal cognitive trends.

Dynamic Difficulty Adjustment Algorithm

Runs locally in `platform.js:getAdaptiveDifficulty()` after each session without sending telemetry to cloud.

Level Progression (Level +1):

Recent 5-game accuracy $\geq 85\%$ with 0 hints used → Smooth transition to higher stimuli count.

Gentle Support (Level -1):

Recent accuracy $< 50\%$ OR hints ≥ 2 → Automatically simplify choice pool to avoid frustration.

Severe Stage Safeguard:

Locks to binary 2-choice tactile buttons, infinite time, and soothing voice guidance.

5-Domain Memory Gap Map

Domain	Sample Games	Clinical Marker Tracked
1. Memory Recall	Tea Estate Memory, Photo Match	Short-term visual retrieval & delayed recognition
2. Spatial Orientation	Brahmaputra Ferry, River Crossing	Topographical navigation & 2D spatial reasoning
3. Language & Naming	Assamese Word Weaver, Rhymes	Semantic fluency & expressive naming
4. Executive & Attention	Odd One Out, Majuli Mask Sorting	Inhibitory control & category shifting
5. Logic & Number Sense	Market Coin Counter, Fish Count	Daily functional arithmetic preservation

AI Avatar Companion: Current Prototype vs. Future Vision

Balancing immediate offline rural feasibility today with bleeding-edge photorealistic neural family avatars tomorrow.

WORKING IN CURRENT BUILD

● ZERO-GPU CLIENT

Current Interactive 3D VRM Pipeline

- ▶ **3-Angle Camera Capture:** `AvatarOnboardingModal.jsx` captures front/left/right photos of family member.
- ▶ **Local Canvas Normalization:** 512×512 normalized texture generated in browser canvas & stored in IndexedDB.
- ▶ **3D Mesh Rigging:** Loaded via Three.js `@pixiv/three-vrm` in `AvatarCanvas.jsx`.
- ▶ **Word-Boundary Lip-Sync:** Speech synthesis events drive real-time mouth morphs (`aa, ih, ou, ee, oh`) via `visemeMapper.js`.

Advantage: Instant 60 FPS on low-cost ₹6,000 tablet, 100% private, zero network requirement.

FUTURE VISION ROADMAP

○ PHASE 2 RESEARCH

Photorealistic Neural Family Avatars

- ▶ **3D Gaussian Splatting / FLAME:** Reconstruct precise 3D head geometry & facial skin geometry from 5-second video scan.
- ▶ **Few-Shot Voice Cloning:** Lightweight XTTS-v2 / Chatterbox integration to clone grandchild or son's comforting voice.
- ▶ **Neural Real-Time Lip-Sync:** LivePortrait / SadTalker diffusion streaming synchronized micro-expressions.
- ▶ **Edge Quantization:** NPU-accelerated ONNX runtime targeting upcoming rural mobile chipsets.

Goal: True photorealistic visual presence of migrating family members for dementia patients.

ASHA Worker Workflow: Multi-Patient Field Operations

Designed for community health workers conducting home visits across Majuli, Bishnupur, and rural districts.

STEP 1: PATIENT SWITCH

Instant Profile Switch

- ▶ ASHA enters secure PIN `CaregiverPinModal.jsx` ('1234') in
- ▶ Selects patient *Biren Das, 72,* or *Memi Devi, 68,* (e.g., *Majuli* *Loktak*).
- ▶ Platform state dynamically switches active baseline and memory anchors.

STEP 2: GUIDED SESSION

Assisted Engagement

- ▶ ASHA worker conducts 15-minute gentle daily check-in + 3 game rounds.
- ▶ "Together Mode" enabled: ASHA guides without penalizing response time.
- ▶ Stars collected and fed directly into the patient's local memory garden.

STEP 3: SYNC & REPORT

Outbox Batch & Export

- ▶ All metrics logged into local FIFO `syncQueue.js` outbox (
- ▶ On reaching PHC with Wi-Fi/4G, outbox auto-flushes in a single batch.
- ▶ Generates printable clinical report PDF for visiting PHC medical officer.

0 KB

Data Needed for Core Sessions

100%

Local Data Encryption & Privacy

< 5s

Batch Auto-Sync Duration

1-Click

Clinical PDF Export for Doctors

Feasibility, Ethics & Clinical Boundaries

Adhering strictly to Indian healthcare privacy, non-clinical ethics, and low-cost hardware feasibility.

HARDWARE FEASIBILITY

Low-Cost Edge Devices

- ▶ Runs smoothly on standard quad-core Android tablets (2GB RAM).
- ▶ PWA install size < 8MB with compressed audio assets.
- ▶ Optional AI backend runs on basic dual-core CPU with Faster Whisper INT8 quantization.

DATA PRIVACY & SECURITY

Zero Cloud Leakage

- ▶ Patient photos, voice samples, and daily mood entries NEVER leave the device.
- ▶ Caregiver access strictly gated by PIN lock.
- ▶ One-click `clearPlatformData()` honoring complete patient data wipe (sovereignty).

ETHICAL BOUNDARY

Non-Diagnostic Guarantee

- ▶ Explicitly marked as an engagement tool, NOT a diagnostic medical device.
- ▶ Never uses scary labels like "Dementia Stage 3" in front of older adults.
- ▶ Reports clearly state: *"Supports care conversations; does not replace neurologist consultation."*

Social Impact & 3-Phase Scalability Roadmap

Scaling from pilot districts in Assam to national primary healthcare integration under Ayushman Bharat.

PHASE 1: CURRENT (SIH GRAND FINALE)

Working Prototype

- ▶ 28 playable cultural games completed.
- ▶ Assamese & English localization verified.
- ▶ PWA offline engine + IndexedDB media vault.
- ▶ Caregiver gap map & PDF export tested.
- ▶ Working 3D VRM avatar with speech lip-sync.

PHASE 2: 3–6 MONTHS

Rural Pilot & Dialects

- ▶ Pilot with 50 ASHA workers in Majuli & Bishnupur districts.
- ▶ Addition of Bodo, Manipuri, and Mizo language packs.
- ▶ Integration of lightweight neural voice cloning on edge devices.
- ▶ Clinical validation studies with regional medical colleges (GMCH).

PHASE 3: 12+ MONTHS

National Scale (ABDM)

- ▶ Integration with Ayushman Bharat Digital Mission (ABDM) ABHA IDs.
- ▶ Photorealistic 3D Gaussian Head Avatars for remote migrant families.
- ▶ Pan-India regional culture packs (Tamil, Bengali, Marathi, Punjabi).
- ▶ MoHFW Tele-MANAS primary care integration.

Empowering Dignified Cognitive Care with Apon Mon

Bridging the gap between modern AI innovation, offline rural resilience, and deep cultural empathy for our elders.

28 Games
Fully Playable

100% Offline
Zero Cloud Dependency

₹0 Server Cost
Edge Architecture

Research & Technical Citations

- ▶ ICMR Dementia India Report (2020): Prevalence & rural healthcare delivery gap.
- ▶ Lancet Commission on Dementia Prevention (2024): Role of cognitive stimulation & socialization.
- ▶ WCAG 2.2 AA Guidelines: Accessible design principles for older adults with cognitive decline.
- ▶ Pixiv 3D VRM & WebGL Standard: Client-side lightweight avatar animation.
- ▶ Smart India Hackathon Benchmarks: KnitKraft, Voco App, DhwaniSarathi architecture patterns.

Thank You! Open for Live Demonstration & Q&A.